



Development of a Small Scale ROV for Research Purposes

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under tutelage of

Tobias Fischer

Acknowledgement of Traditional Owners

The Queensland University of Technology (QUT) acknowledges the Turrbal and Yugara, as the First Nations owners of the lands where QUT now stands. We pay respect to their Elders, lores, customs and creation spirits. We recognise that these lands have always been places of teaching, research and learning.

QUT acknowledges the important role Aboriginal and Torres Strait Islander people play within the QUT community.

Project Overview

Task Description:

This project involves designing, building, and programming a miniature **autonomous** underwater vehicle to serve as a demonstrator for **educational outreach** and **testing platform for underwater robotics research**. The objective of this project is to create a **small-scale** submersible capable of controlled diving, surfacing, and maneuvering in test tanks or demonstration pools.

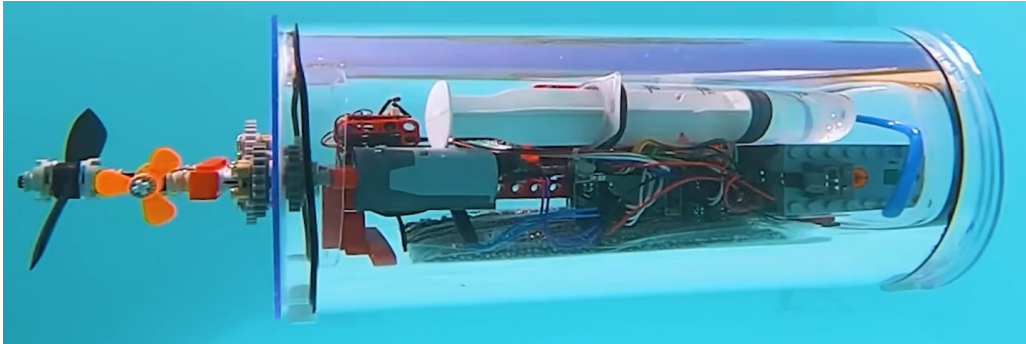


Photo which inspired the project shared by Supervisor

The final demonstrator should include:

1. Controlled Maneuverability: Achieving stable, controlled diving and movement using a thruster configuration.
2. Help complement ongoing research with our BlueROV ROV by providing a cost-effective platform
3. Help guide algorithm development, open day demonstrations, and initial machine learning pipeline testing.

Project Breakdown

Research Topic: Development of a Small Scale ROV for Research Purposes

Research Question: How can a modular, small-scale ROV platform be developed using primarily Commercial Off-The-Shelf components to create a cost-effective and adaptable tool for undergraduate-level robotics education and system testing?

Research Problem: Currently the BlueROV2 ROV that QUT owns is too large and use requires additional steps including

- QUT pool approval and/or offsite setup
- Difficult manual handling
- This larger platform can be overkill for small tests

This project aims to design a smaller solution which directly solves these problems



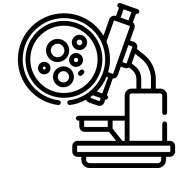
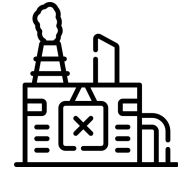
Planned Project Outcomes and Broader Impact

Expected Outcomes:

- A fully assembled and operational small-scale underwater ROV.
- Complete Computer-Aided Design (CAD) drawings and a detailed Bill of Materials (BOM) for the final design.
- Control system software with a camera view.
- Autonomous control with plans for AI-ML Integrations
- A final report detailing the design methodology, validation, and performance results.

Benefits & Impact:

- Academic & Research: Creates a versatile, low-cost testbed for QUT demonstrations and future student projects, enabling hands-on learning in marine robotics, control systems, and sensor integration.
- Future Innovation: Provides an adaptable platform for testing novel control pipelines, serving as a foundation for more advanced research.
- Sustainability: Delivers a long-term, economically sustainable asset through a modular and repairable design using COTS components, maximizing the return on investment and ensuring knowledge transfer to future teams.



Literature Review

~60–70 papers were identified, 3 Key
Papers discussed

Comparative Analysis of Key ROV Literature

1. Modular Observation ROV (BlueROV2)

ROV Focus: Focusing on surveillance, inspection, and mapping.

Philosophy: Modularity, stabilized frame, and 5 DOF maneuverability.

Jothikrishna et al. (2023)

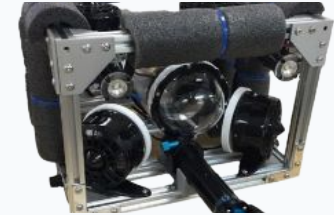


2. Hybrid ROV - Capstone Project

ROV Focus: Educational platform for shallow channel survey and MATE.

Philosophy: Dual ROV/AUV modes and autonomous depth profiling.

Hur, B., & Alvi, M. (2022)



3. Hybrid AUV/ROV for Aquaculture

ROV Focus: Kalypso, designed for Mediterranean aquaculture inspection.

Philosophy: Cost-effective 3D printing and autonomous net inspection.

Manos et al. (2024)



3 Phase Approach

Phase 1: Research and Design

Research: Research components and materials used in current solutions

Development: Iteratively design a first stage solution that meets research outcomes

Phase 2: Simulation and Prototyping

Simulation: Produce a digital twin that allows simulation of buoyancy and motion in ros2 using Solidworks documentation

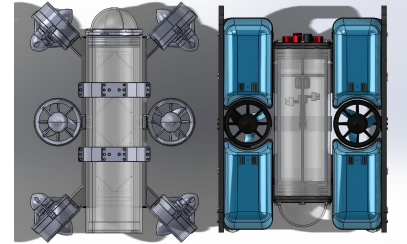
Prototyping: Develop a second stage refined design based off simulation results

Phase 3: Reporting, Testing & Refinement

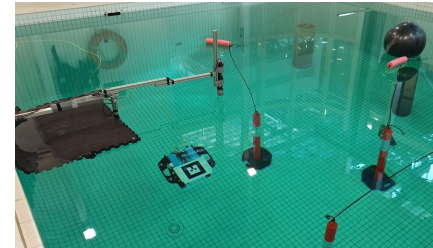
Testing: Conduct waterproof tests, Manual motion tests and possibly autonomous identification tests

Refinement: Use test results to refine a third stage solution that has evidence meeting research outcomes.

Reporting: Document all outcomes and finalize outcomes into a accessible package(Github repo)



ROS 2TM



Project Management & Adaptation

Initial Project Plan

Semester 1: Phase 1 focus on literature review, component research, and CAD design iterations.

Semester 2: Phase 2 STL/ROS2 export & buoyancy simulation. Phase 3 physical prototype construction and testing.

Roadblocks

- Small scale design complexities during fabrication
- Unforeseen design constraints and integration issues
- High powered battery safety and usage management

Required Plan Adaptations

Team Expansion:

Onboarded Brad Edwards (Simulation Engineer) to parallelize the digital twin development. Moving the project from solo to small team.



Environment Change:

Pivot from QUT Pool to private pool testing to accelerate physical trials.



Scope Refinement:

Re-evaluate the scope of the project and remove non-critical elements to ensure key deliverables were achieved



Risk Assessment and Health and safety considerations



Risk Assessment : 18339 (Pending Endorsement)

Customer Details:-

Owner: Josh Hecke (joshua.hecke@connect.qut.edu.au)

Review: Yes

Email: joshua.hecke@connect.qut.edu.au

Endorsement: Yes

Assessment Type: Plant/Equipment

MAPS ID:

Division/Faculty: Faculty of Engineering

QUT Team: School of Electrical Engineering & Robotics

School/Depart: School of Electrical Engineering & Robotics

QUT SubTeam:

Start Date: 8/25/2025

Est End Date: 6/25/2026

Title: Design of small scale Remotely Operated Vehicle - Underwater Robot

Description: This is an risk assessment to support the design of a new small scale ROV on campus.

The primary risks relate to electrical safety during development, manual handling, operation, electrical safety including charging and packing and storage for travel.

Only Tobias Fischer, Joshua Hecke and Scarlet Raine will be able to operate this device during development and testing, due to its experimental nature. Other team members can operate this vehicle after contacting said operators and receiving a induction.

Main Risks-

5200mAh 14v LiPo Battery Use and Handling

Operation and Handling around water

IF USED

Safe use and Storage of Dichtol Waterproofing

Ensure-

S9 lab techs are aware of LiPo presence if battery is on campus for safe storage and charging.

In addition to this risk assessment, Approval of any commercial areas(i.e qut pool) needs to be organized with the appropriate body accordingly.

Location Details:-

Campus	Building	Floor	Room	Additional Info
Off Campus				These are operated on-water in the field at approved locations(A key example would be QUT Pool or Personal Pool)
Gardens Point	S Block	Level 11	1145-Post-Graduate Lab.	Mobile Robots Lab Soldering Station

Primary Hazards identified in this project

-Electrical and Tool safety during prototyping

-Water Ingress & Electrical Shorts.

-High-Energy LiPo Battery thermal runaway.

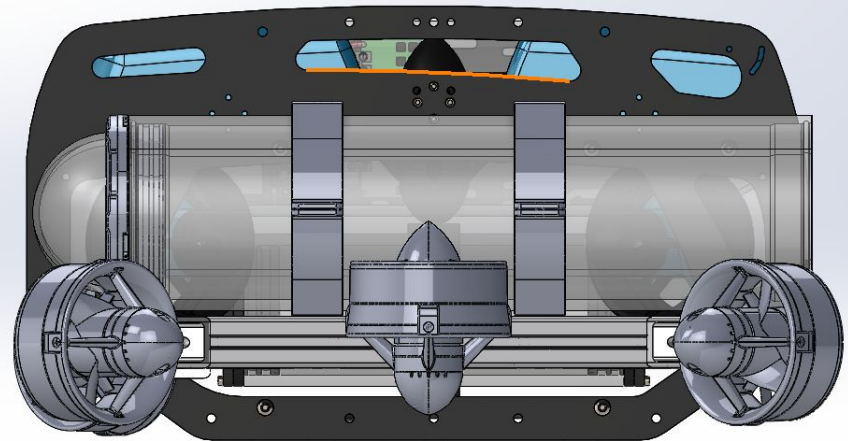
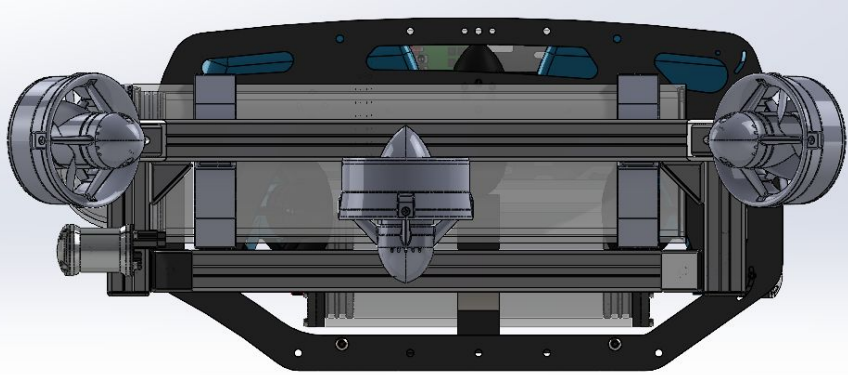
-Manual Handling and Water Safety

Detailed risk **Mitigation** and **Removal** performed in Risk Assessment

Methodology

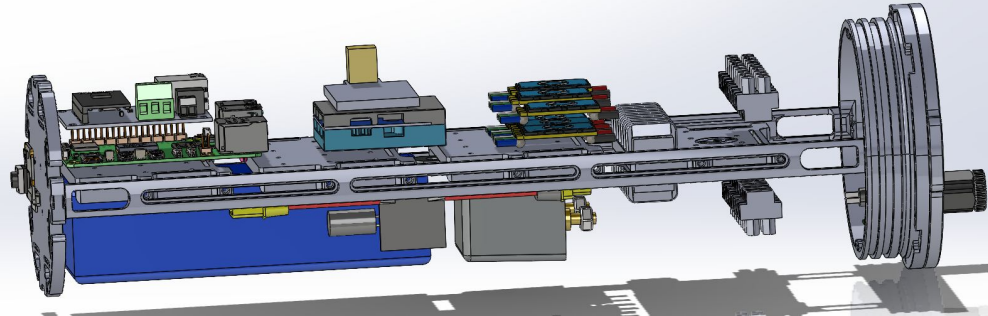
Phase 1 – Phase 2

Design Iterations #1 and #1.1



Stakeholder Feedback:

- Design is too large!
- Consider use case again as this design misses most criteria
- Fabrication Request Denied



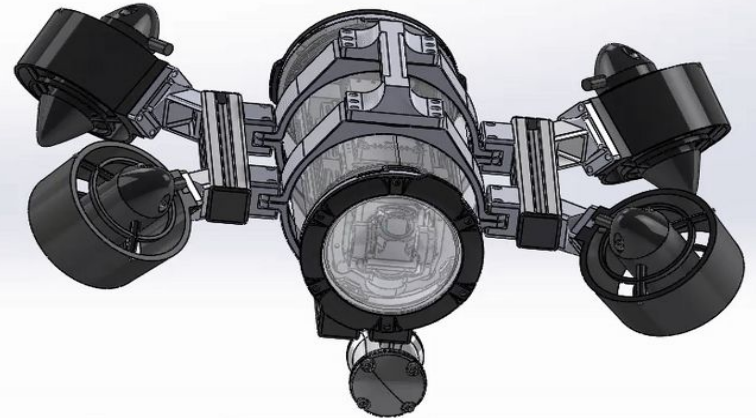
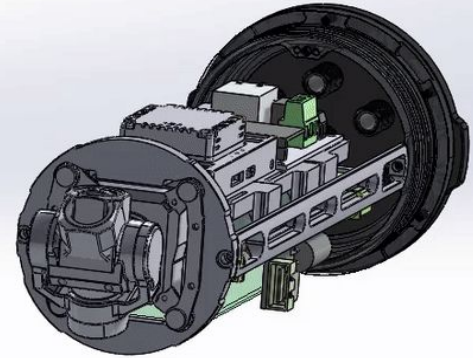
Design Iterations #2 and #2.1

Stakeholder Feedback:

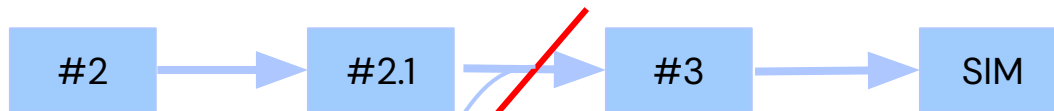
Request for design to be fabricated, noting introduced effect of downsizing

- Underactuation
- Smaller Battery time

-Fabrication request Granted as the design met most criteria!



Roadblocks



#2/#2.1 Challenges:

- Experimental 45-degree vectored thrusters lacked empirical backing.
- Severe internal spatial conflicts with wiring harnesses and the real size of selected components
- No real plan for mounting additional buoyancy foam
- Camera planned for #2 was no longer sourceable

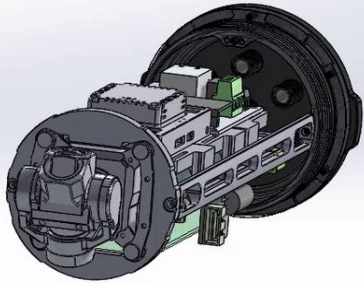
#3 Engineering Solutions:

- Reverted to traditional 4-thruster layout (2 vertical, 2 horizontal) for proven hydrodynamic stability.
- Designed custom 3D-printed PLA internal chassis to maximize the 3-inch pressure hull volume.
- Downsized from 5200mAh LiPo to a compact 2,200mAh LiPo battery.
- Swapped obsolete Px4Flow camera for Z-1 Mini Gimbal and added an external handle.

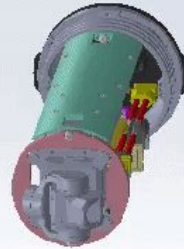
Final Iteration #3 Full Assembly



Final Iteration #3 Internal Mounting



#2 for Comparison



Fabrication Simulation Testing

Phase 3

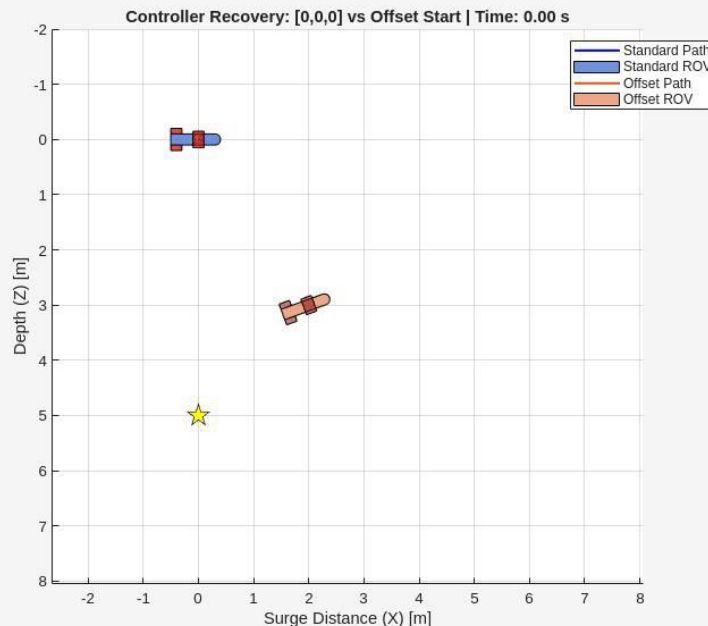
Underactuation Simulations

Matlab Underactuation Simulation

Underactuated Dynamics: MATLAB LQI study proved 3-DOF planar model stability despite severe non-linear drag and lack of independent pitch control.

Important Note:

This study was done to support the selection for underactuation, variables used in this study do not match the SubbyROV and capabilities for position measurements are a requirement for the LQI controller. This work can also be used to support future control and modeling



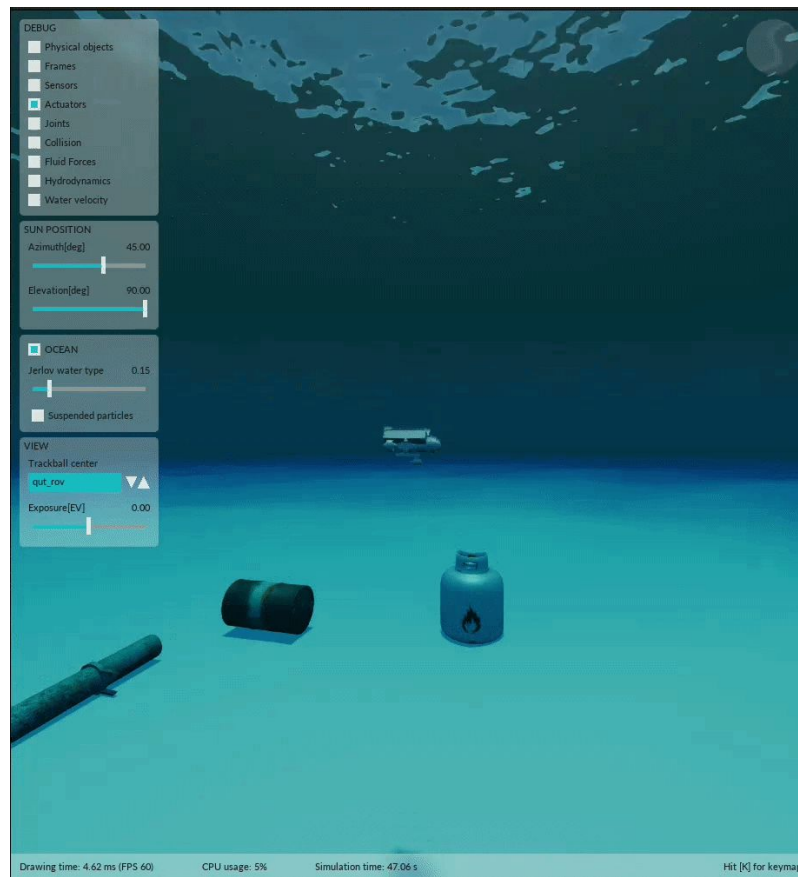
Stonefish Simulations

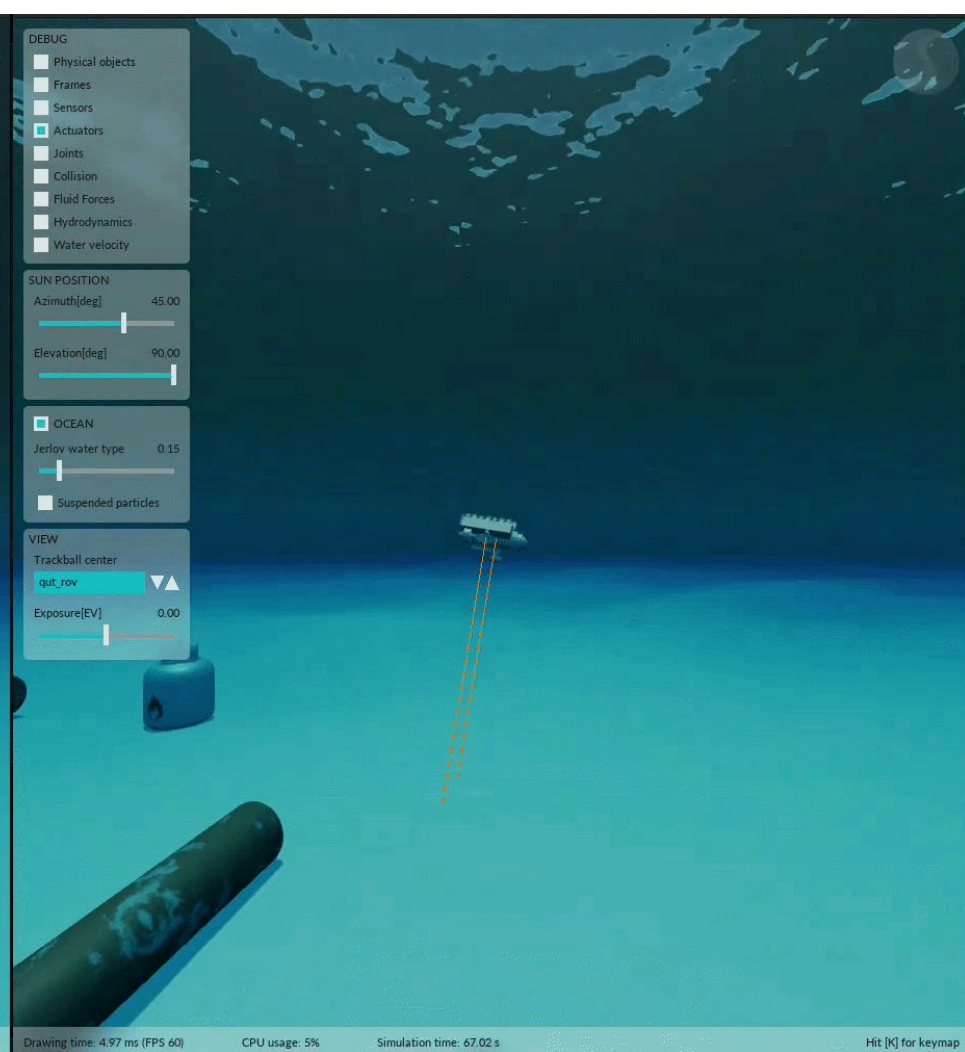
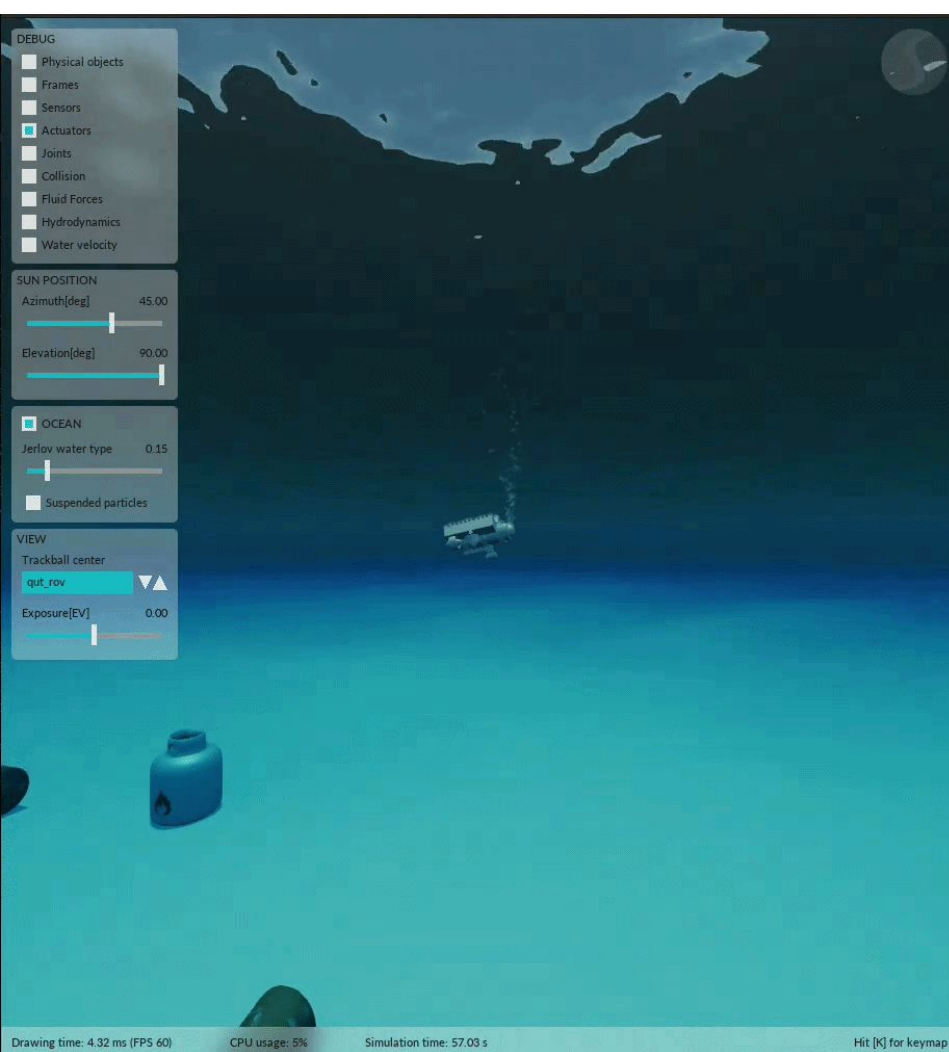
Spearhead by Brad Edwards

Digital Twin: Exported SolidWorks STL mesh to custom designed Stonefish ROS2 environment.

Buoyancy Calibration: Initial simulations showed Center of Buoyancy was too far forward (nose-up pitch). Corrected using a Iterative testing process before fabrication

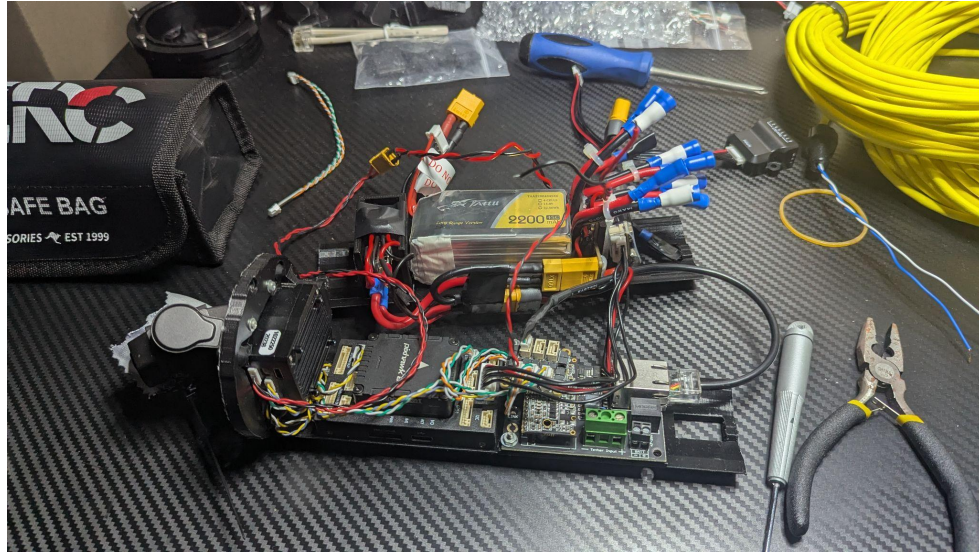
Dynamic Simulation: Simulated simple thruster inputs to view the designs transient and steady state response



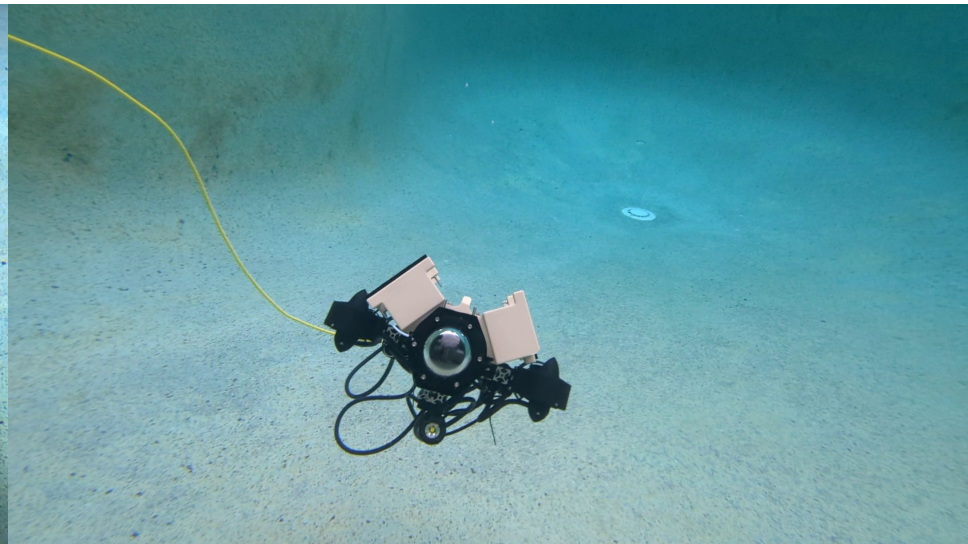
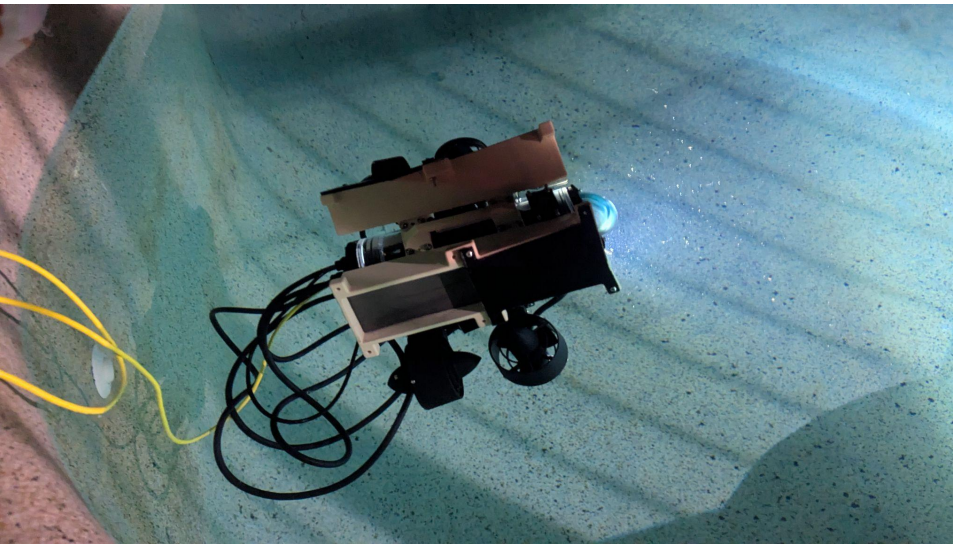




Physical Assembly Internals



Real World Testing



Test 1: Initial demo and operations test

Objective: To test all basic functions in operating Environment

Results: Qualitative assessment of Digital Twin and real world Differences

Test 2: Manual Control

Objective: To mimic simple inputs that can be/were replicated in the Stonefish simulator.

Results: Qualitative Results in the form of Videos.

Test 1: [VideoLink](#) Test 2A: [VideoLink](#) Test 2B: [VideoLink](#)

Limitations and Significance

Significance / Strengths

- **Effective & Modular:** Successfully delivered a highly adaptable, low-cost (~\$2884 AUD) testbed utilizing Commercial Off-The-Shelf (COTS) and custom 3D-printed components.
- **Simulation Validation:** Capable of qualitatively evaluating complex underactuated dynamics, effectively validating the custom Stonefish digital twin.
- **Research Foundation:** Establishes a robust, open-source hardware and software baseline for future autonomous navigation and machine learning research.

Limitations

- **Underactuated Dynamics:** The 4-thruster configuration lacks independent pitch control, leading to pitch angle oscillations during surge movements without additional control.
- **External Sensing:** The absence of positional sensors or a Doppler Velocity Log (DVL) **restricts immediate** advanced autonomous operations.
- **Material Durability:** Printed components are excellent for rapid prototyping but are prone to warping and splitting, making them unsuitable for long-term submersion.
- **Operational Capacity:** Spatial constraints required downsizing to a 2200mAh battery, limiting continuous operational time to approximately 10 minutes.

Conclusion and Future Directions

Outcomes:

- **Successful Prototyping:** Delivered a functional, low-cost ROV platform capable of manual teleoperation and academic demonstration.
- **Physical Validation:** Overcame significant mechanical and electrical spatial constraints, successfully validating waterproof integrity and basic maneuverability in a pool environment.
- **Digital Twin Implementation:** Established a robust Stonefish simulation environment that dynamically models the vehicle's complex underactuated setup
- **Open-Source Access:** All findings and files produced by work can be located in the projects Github Repository

Future Work:

- **Sensor Expansion:** Integration of external state-estimation sensors, such as a Doppler Velocity Log (DVL) and high-resolution depth sensors, to enable accurate localization.
- **Thruster Redesign:** Using current work as a starting point the design can be fine tuned with highly customised parts to hold 6 thrusters
- **Advanced Control Systems:** Development and deployment of robust control laws to manage the ROV's highly non-linear, underactuated dynamics.
- **Machine Learning Autonomy:** Utilizing the onboard IP camera to implement ML pipelines and computer vision algorithms for object detection, path following, and fully autonomous navigation.

Any Questions?

References

Hur, B., & Alvi, M. (2022). Educational Small Scale Underwater Robot Development via a Capstone Project in Engineering Technology. ASEE Annual Conference and Exposition, Conference Proceedings.

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This water was so cold but the light demo too cool to miss